

CROSS-DOMAIN VOICE ACTIVITY DETECTION ON HEALTHY AND NEURODEGENERATIVE SPEECH ACROSS NARRATIVE AND DIADOCHOKINETIC TASKS

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ABSTRACT

Voice activity detection (VAD) underlies automated speech analysis, from transcription to clinical monitoring, yet VAD models are developed primarily on speech from healthy speakers. Neurodegenerative diseases can produce speech with reduced speed, precision, and continuity. Such speech remains underrepresented in VAD evaluation. We benchmark two neural VAD architectures, NAS-VAD and Self-Attentive VAD, trained on public noisy-speech corpora, fine-tuned on clinical recordings from the Ontario Neurodegenerative Disease Research Initiative (ONDRI), and evaluated on held-out clinical and healthy-control ONDRI recordings across narrative and diadochokinetic (DDK) tasks. Across tasks and labelling definitions, F2 scores ranged from 0.89-0.97, but task- and label-dependent errors remained. Precision fell to 0.74-0.75 on DDK speech, consistent with over-detection across brief inter-syllable silences, while both models frequently missed breaths and vocal gestures under the broader all-vocalizations definition. These patterns may reflect biases introduced by off-the-shelf pseudo-labels and highlight the importance of clinically grounded VAD evaluation in neurodegenerative speech applications.

Index Terms— voice activity detection, neurodegenerative speech, clinical speech monitoring, cross-domain evaluation, pseudo-label bias

1. INTRODUCTION

Voice activity detection (VAD) identifies the portions of an audio recording that contain speech and is a foundational preprocessing step in automated speech analysis. Its output determines which signal segments are passed to downstream systems for transcription and extraction of acoustic and temporal measures, such as speaking rate, pause duration, and speech-to-silence ratio. Errors at this stage can therefore propagate throughout the analysis pipeline: false-positive detections may classify silence or noise as speech, whereas false-negative detections may remove weak, brief, or atypical vocaliza-

tions. These errors are particularly consequential in clinical applications, where changes in speech timing and continuity may themselves provide markers of neurological disease or disease progression [3]. Accurate VAD is therefore important not only for downstream processing but also for the validity of speech-based clinical measures.

Most contemporary VAD systems are trained and evaluated using recordings of healthy speakers producing conventional connected speech [4, 5]. Neurodegenerative disease, however, can alter vocal intensity, articulation, speaking rate, pause structure, respiratory support, and the production of non-word vocalizations. VAD performance may also depend on the speech task being analyzed. Narrative speech contains variable utterance and pause durations, whereas diadochokinetic (DDK) tasks consist of rapid syllable repetitions separated by brief and sometimes irregular intervals. Prior work has shown that VAD configurations optimized for one clinical population may not transfer reliably to another [6]. Similarly, strong performance on healthy connected-speech benchmarks cannot be assumed to generalize either to neurodegenerative speech or across clinically used speech tasks.

Despite the increasing use of automated speech measures in clinical research [7], modern neural VAD architectures have rarely been evaluated across both healthy and neurodegenerative speech using task-specific, manually corrected ground truth. Most public VAD datasets also represent a comparatively narrow range of speech behaviours, leaving it uncertain how models trained on these resources perform when confronted with the altered timing, intensity, and vocal behaviours that may occur in neurodegenerative disease.

We address this gap by evaluating two reproducible neural architectures, NAS-VAD [4] and Self-Attentive VAD [5], across narrative and DDK recordings from healthy speakers and participants with neurodegenerative disease. Both models were trained on public noisy-speech corpora and subsequently fine-tuned using clinical recordings labelled by an off-the-shelf VAD system. We then evaluated the models against manually corrected annotations. Specifically, we examine: (1) whether performance generalizes across narrative and DDK tasks [8]; (2) whether performance differs between healthy and neurodegenerative cohorts; and (3) how performance changes when speech is defined narrowly as word and non-word productions or more broadly to include audible breaths and vocal gestures. This design evaluates overall VAD performance and examines whether the observed error patterns are consistent with systematic biases in the automatically generated pseudo-labels used for fine-tuning.

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2. METHODOLOGY

2.1. Literature Review and Public Datasets

We reviewed recent VAD literature and selected two reproducible neural VAD systems with publicly available implementations: NAS-VAD, a compact architecture identified through neural architecture search, and Self-Attentive VAD, which uses self-attention and a boosted prediction module. The models provide architecturally distinct approaches and have demonstrated strong VAD performance.

For training, we selected three widely used public VAD datasets: VOiCES [9], VoiceBank + DEMAND [10], and MS-SNSD [11]. Together they span a wide range of room conditions, noise types, and signal-to-noise ratios (SNRs), letting us train under acoustically challenging conditions; dataset-specific details and splits are given in Section 3.

2.2. ONDRI Clinical and Healthy-Control Cohorts

We used clinical and healthy-control recordings from the Ontario Neurodegenerative Disease Research Initiative (ONDRI) [1]. The clinical cohort included participants with neurologist-confirmed diagnoses of Alzheimer’s disease/mild cognitive impairment (AD/MCI), amyotrophic lateral sclerosis (ALS), frontotemporal dementia (FTD), Parkinson’s disease (PD), or vascular cognitive impairment (VCI). The similar-aged healthy-control recordings were collected specifically as a comparison group for ONDRI and also form part of the larger BEAM dataset. We refer to these healthy controls as the BEAM cohort throughout the paper. From the ONDRI clinical and healthy-control cohorts, we selected two categories of speech tasks: diadochokinetic (DDK) and narrative speech. DDK tasks are designed to assess motor control of speech and include two types of activities: alternating motion rate (AMR), which involves repeating single syllables such as ‘puh,’ ‘tuh,’ or ‘kuh,’ and sequential motion rate (SMR), which involves repeating a sequence such as ‘puh-tuh-kuh’ or ‘buttercup.’ Narrative tasks (SeqStorT1, SeqStorT2A) involve connected, natural speech produced while generating a story to a set of picture stimuli [12]. We selected participants from both datasets to compare healthy and neurodegenerative speech under matched recording conditions and speech activities. Participants were included if they completed three AMR tasks, one of the two SMR tasks, and two narrative tasks. With the exception of the AD/MCI cohort, each selected participant had recordings for all six activities, and TextGrid annotations were already available for the AMR and SMR tasks. Only the two narrative tasks were available for participants with AD/MCI. When narrative TextGrids were unavailable for participants in the representative subset, we generated them using the pipeline described in Section 2.3 and then manually corrected the annotations.

2.3. Narrative TextGrid Generation Pipeline

We built a pipeline to generate orthographic transcripts for narrative speech recordings that did not already have transcriptions. Audio recordings were then resampled to 16 kHz, 16-bit PCM and transcribed using Whisper large-v3 [13]. Transcripts were aligned to the audio at the word level using the Montreal Forced Aligner (MFA, v3.3.9) [14]. The MFA output included two annotation tiers, “words” and “phonemes”, which were collapsed into a single “vocalizations” tier for the purposes of this study. These TextGrid files were then set aside for manual, word-level annotation using Praat (v6.4.67) [15].

2.4. Ground Truth Label Generation Procedure

DDK TextGrid files were manually annotated in Praat by two independent raters, and their labels compared for discrepancies. Discrepancies between labels were resolved through the consensus of a third rater. Narrative TextGrid files were manually corrected in Praat using a standardized labelling procedure but were not independently duplicated, which may introduce variability in annotations. The narrative TextGrid procedure classified annotated vocal events into four categories, each marked in the TextGrid files with a ‘\$’ prefix to distinguish annotation tags from transcribed speech content: ‘\$word’, ‘\$non-word’, ‘\$vocal-gesture’, and ‘\$audible-breath’. The ‘\$word’ label marks any spoken word or uninterrupted sequence of words. The ‘\$non-word’ label marks filler words, partial words, and phonological fragments that are not intelligible. The ‘\$vocal-gesture’ label captures vocal activity outside these categories, such as coughing or laughter. The ‘\$audible-breath’ label marks any audible breathing. All four labels were applied as time-aligned intervals within a single “vocalizations” tier. The corresponding audio served as input to the VAD models, and the hand-corrected TextGrid annotations served as the evaluation reference. Figure 1 shows an example segment annotated with all four categories.

2.5. Training Method

We used Silero-VAD [16] with its default segment-detection parameters to generate pseudo-labels for both the public training data and the ONDRI data held out for fine-tuning. These segment-level labels were expanded first into sample-level binary speech masks, then into frame-level binary arrays for each recording. Frame-level labels were assigned using a fixed majority rule: within a 25 ms window with a 10 ms hop, a frame was labelled as speech if more than half of its samples fell inside a speech segment. For held-out narrative evaluation recordings, we converted the manually corrected TextGrid files into frame-level arrays using two labelling schemes. The first, “Word/Non-Word” (WNW) labelling, treats only the ‘\$word’ and ‘\$non-word’ categories as speech. The second, “All-Vocalizations” (AV) labelling, treats all four annotation categories as speech. Comparing these two schemes lets us test how model performance changes when vocal activity outside ‘\$word’ and ‘\$non-word’, such as breathing or vocal gestures, is or is not counted as speech. To generate model inputs, we computed spectrograms using the Short-Time Fourier Transform (STFT) with a 400-sample window, a 160-sample hop length, and a 16 kHz sampling rate. Each resulting spectrogram, containing the raw STFT magnitude, was paired with its corresponding binary label array as input to the model.

3. EXPERIMENTAL SETUP

3.1. NAS-VAD repository

We used the public code repository for NAS-VAD [17], which also includes the Self-Attentive VAD baseline, so that both architectures could be evaluated under the same framework. After training on public data, each model was fine-tuned on the ONDRI fine-tuning data using two separate procedures. The first used the repository’s default hyperparameters, which we refer to as the baseline configuration. The second used hyperparameters identified through an Optuna-based search [18]. We kept the repository’s original methods for architecture-specific input construction, including context window sampling, log-mel feature extraction, and data augmentation. We did, however, extend the training code to support Optuna-

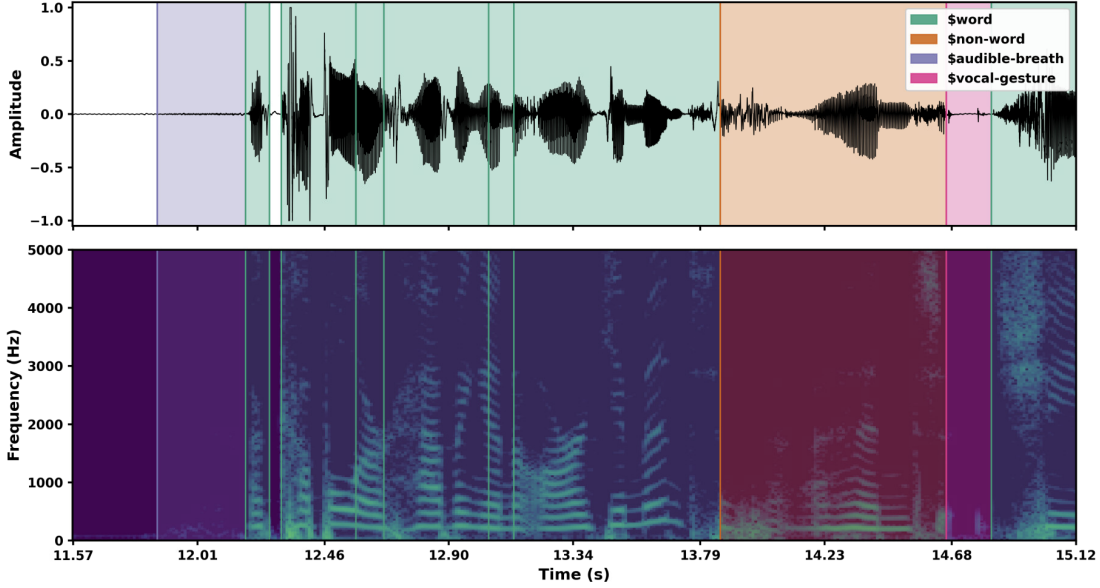


Fig. 1. Hand-corrected narrative segment showing all four annotation categories across the waveform and spectrogram.

based hyperparameter search, dataset- and cohort-specific configurations, and F2-based checkpoint selection, replacing the original code’s default criterion of lowest validation loss. Our extended training and evaluation pipeline, along with participant allocation details, the full hyperparameter search space, and additional validation and performance metrics, is available at <https://github.com/ICCARe-Hub/ND-VAD-2026>.

3.2. Public Datasets Training and Validation Splits

We combined public training data from VOiCES, VoiceBank + DEMAND, and MS-SNSD into speaker-disjoint training and validation sets. For VOiCES, we used the official development kit, including only close-microphone recordings, babble or television noise types, across the training, validation, and testing splits. For VoiceBank + DEMAND, 26 of the 28 available speakers were used for training, with the remaining two held out for validation; all noise types and SNR levels from the dataset were retained, and the official two-speaker test partition was used for testing. We used MS-SNSD to further expand the training and validation data. The MS-SNSD dataset provided clean speech from the Edinburgh 56-speaker dataset [19], mixed with babble and air-conditioning noise at 20 dB and 25 dB SNR, extending the noise conditions already present in VoiceBank + DEMAND to additional, higher-SNR levels.

3.3. ONDRI Training and Validation Splits

The ONDRI clinical subset contained 514 participants and 2,528 recordings. From this subset, we constructed a balanced representative pool of 12 participants per diagnostic cohort. The remaining 454 clinical participants were used for fine-tuning and divided into 364 training and 90 validation speakers. The ONDRI healthy controls, referred to as the BEAM healthy-control cohort, were reserved for held-out evaluation and were not used for fine-tuning. Manually corrected TextGrid files were used as reference annotations for all model evaluations. Model performance was assessed on ten held-out participants from each of the AD/MCI, PD, and BEAM healthy-

control cohorts using one narrative TextGrid and four DDK TextGrid files per participant. The remaining representative-pool participants were reserved for future annotation and were not used in the present evaluation.

3.4. Hyperparameter Fine-Tuned Setting and Metrics

In an additional experiment, we ran fifty Optuna trials per model using the default sampler, searching nine hyperparameters spanning training controls (batch size, epochs, learning rate, weight decay, gradient clipping, early stopping), model-specific regularization (dropout, drop path), and the loss function (BCE, weighted BCE, or focal loss with $\gamma=2$ [20]). Each trial trained and validated on the 454 fine-tuning speakers, retaining the highest-validation-F2 checkpoint. Candidate trials were ranked primarily by validation F2. When trials had F2 scores within 0.001 of one another (i.e. $|F2_1 - F2_2| \leq 0.001$), the remaining validation metrics were considered when selecting the final configuration. The selected configuration per architecture, baseline or optimized, was then evaluated on the ground-truth TextGrid files. We report the area under the receiver operating characteristic curve (ROC AUC), accuracy, precision, recall, F1, and F2, with all metrics except ROC AUC computed at a fixed threshold of 0.5. F2 serves as our primary comparison metric, weighting recall over precision to reflect the clinical priority of minimizing missed speech. All metrics were computed after pooling frames across all recordings. Frame-level predictions were generated at inference using the repository’s sliding-window procedure.

4. RESULTS

4.1. Validation Results

Based on validation F2 and performance across secondary metrics, we selected the NAS-VAD baseline fine-tuned model and the Self-Attentive VAD Trial 2 checkpoint; complete validation results and hyperparameter configurations are available in the repository.

Table 1. Global performance across the narrative tasks under WNW labelling and AV labelling, and DDK speech tasks.

Task	Model	AUC	Acc.	Prec.	Rec.	F1	F2
Narrative (WNW)	NAS-VAD	.984	.927	.908	.983	.944	.967
	Self-Attentive	.966	.880	.855	.975	.911	.948
Narrative (AV)	NAS-VAD	.946	.871	.941	.878	.909	.890
	Self-Attentive	.930	.852	.905	.890	.898	.893
DDK	NAS-VAD	.960	.884	.740	.9997	.850	.934
	Self-Attentive	.988	.890	.750	.999	.857	.937

AUC: ROC AUC; Acc.: accuracy; Prec.: precision; Rec.: recall.

Table 2. Per-cohort performance on the narrative (WNW labelling) and DDK speech tasks.

Task	Model	Cohort	AUC	Acc.	Prec.	Rec.	F1	F2
Narrative	NAS-VAD	ADMCI	.978	.918	.897	.986	.939	.967
		PD	.978	.917	.899	.970	.933	.955
		BEAM	.992	.941	.924	.990	.956	.976
	Self-Attentive	ADMCI	.950	.863	.843	.966	.900	.939
		PD	.958	.877	.851	.962	.903	.938
		BEAM	.983	.895	.866	.991	.924	.963
DDK	NAS-VAD	PD	.959	.877	.735	.9998	.847	.933
		BEAM	.962	.890	.744	.9997	.853	.935
	Self-Attentive	PD	.988	.884	.746	.9999	.854	.936
		BEAM	.988	.896	.755	.998	.860	.938

AUC: ROC AUC; Acc.: accuracy; Prec.: precision; Rec.: recall.

4.2. Ground-Truth Data Evaluation

After selecting the final checkpoint for each architecture, we evaluated both models on the held-out participants across the AD/MCI, PD and BEAM healthy-control cohorts for the narrative and DDK speech tasks. The narrative evaluation used the WNW labelling approach. Global and per-cohort performance for both models, across both speech task categories, is reported in Table 1 and Table 2.

4.2.1. Global DDK & Narrative results

On DDK, both models achieved near-perfect recall but lower precision, indicating over-detection of speech. Figure 2 illustrates detections spanning brief inter-syllable silences. Self-Attentive VAD performed marginally better than NAS-VAD on most DDK metrics. On narrative speech, NAS-VAD performed better across all metrics and both models achieved substantially higher precision than on DDK. Thus, recall remained high across tasks, but precision was task dependent.

4.2.2. Per-cohort DDK & Narrative Results

On narrative speech, frame-level performance was modestly higher in BEAM healthy controls than in the AD/MCI and PD cohorts (Table 2). On DDK, Self-Attentive VAD performed marginally better than NAS-VAD in PD, while the models performed similarly in

BEAM healthy controls. Both models had lower F1 and F2 on DDK than on narrative speech, primarily because of lower precision.

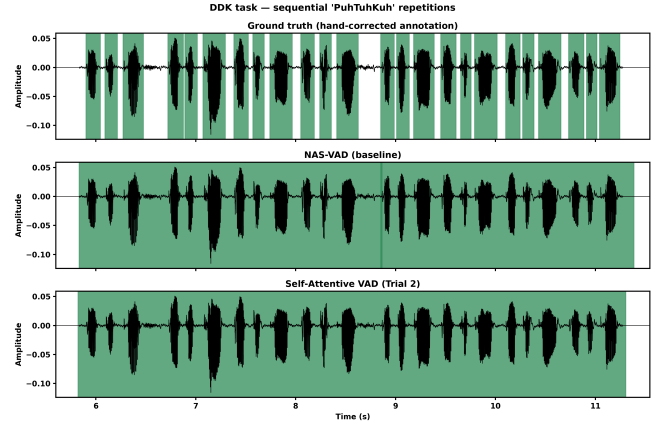


Fig. 2. A DDK segment: ground-truth speech regions compared with both models' frame-level predictions

4.3. WNW vs AV labels for the narrative task

We also evaluated the selected models under AV labelling, comparing performance on the same narrative recordings under two definitions of speech. This comparison is motivated by the difficulty of standardizing the classification of vocal activity outside the '\$word' and '\$non-word' categories, both within and across neurodegenerative diseases. Global performance under both approaches is reported in Table 1; miss rate, false alarm rate, and raw prediction counts are provided in our repository. Relabeling '\$vocal-gesture' and '\$audible-breath' as speech added 22,947 positive frames to the ground-truth data. NAS-VAD and Self-Attentive VAD missed 77.8% and 64.0% of these added frames respectively, with false negatives rising from 2,454 to 20,300 and from 3,585 to 18,270. This increase in false negatives is consistent with pseudo-label bias: Silero VAD, used with its default parameters, may not be calibrated to detect breaths and other non-lexical vocal activity. Accurately classifying distinct vocalization types is highly relevant for clinical use and downstream analysis, making this a useful direction for future work.

5. CONCLUSION

This study evaluated NAS-VAD and Self-Attentive VAD on held-out clinical and healthy-control recordings from ONDRI across narrative and DDK tasks. Both models achieved high recall and ROC AUC, but precision was substantially lower for DDK speech. They also frequently missed breaths and vocal gestures when these events were included in the definition of vocal activity. These task- and label-dependent errors may reflect biases introduced by the off-the-shelf pseudo-labels used for fine-tuning. Because false-positive detections can distort clinically relevant measures such as pause duration, speech duration, and speech-to-silence ratio, high recall alone is insufficient for clinical speech analysis. Future work should fine-tune models directly on manually labelled clinical data and evaluate performance across broader recording environments, speech tasks, and neurodegenerative conditions.

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